

Code-Layout, Readability and Reusability

Date	01 July 2026
Team ID	
Project Name	Smart Lender Loan Eligibility Prediction System
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	4-space indentation maintained across Python files
2	Proper File Structure	Files and folders are logically organized	Yes	Separate folders for models, static, templates, and routes
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables like loan_amount, credit_history are descriptive
4	Function / Method Names	Functions are descriptively named	Yes	Functions like preprocess_data(), predict_outcome()
5	Code Comments	Inline and block comments explain logic	Partial	Comments present in ML scripts, but need more in Flask routes
6	Modular Design	Code is split into reusable functions/modules	Yes	ML pipeline modularized into preprocessing, training, prediction
7	No Redundant Code	Duplicate or unused code is removed	Yes	Cleaned after model selection (only XGBoost retained for deployment)

8	Error Handling	Exceptions and errors are handled gracefully	Partial	Basic try-except blocks present, but logging can be improved
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Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Data Preprocessing Module	Python (pandas, numpy)	Used across all ML models	High
2	Model Training Script	Python (scikit-learn, XGBoost)	Reused for multiple algorithms	Medium
3	Prediction Function	Python (Flask API)	Called by web app for real-time predictions	High
4	Authentication Module	Flask + JWT	Reused across login and admin dashboard	Medium
5	Reporting Dashboard	Flask + matplotlib/seaborn	Generates analytics for officers/analysts	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized with clear separation of concerns
Readability	4	Mostly clear, but comments can be expanded
Reusability	4	Preprocessing and prediction modules highly reusable
Documentation / Comments	3	Needs more detailed docstrings and API documentation
Overall Score	4.0 / 5	Strong structure and modularity; minor improvements in comments and error handling recommended